CAD input



Saved source solid

Instruction

Throat radius
58.7 -> 40.0 mm

Code + request

Frozen Astra

Astra

Generate line patch



Initial candidate

Offline IoU: 0.7936

Trained verifier

Source + request
+ candidate

Qwen 1.5B

LoRA rank 8

>_ Execute CAD
Valid solid

Learned verdict: REJECT

One repair

Same Astra

Answer + rejection
+ execution feedback

JSON line patch
One bounded retry



Repaired candidate

Decision

Same verifier
ACCEPT

Use repair
IoU = 1.0*

If rejected:
keep original

Throat sections: inner radius = 40 mm

Initial: outer 63.2 mm

Repair: outer 81.9 mm

Completed verifier GRPO (training only)

Saved Astra answers -> 8 action trajectories -> verdict reward -> LoRA

Reference geometry supplies training labels, never inference observations

Student-editor pipeline: SFT complete; GRPO not yet completed

Astra examples -> SFT Qwen -> generated CAD -> geometry reward -> editor GRPO

All tasks: 89/124 -> 91/124

2 recovered / 0 regressed